

CHAPTER 7: ELECTRICAL TRANSMISSION POWER LINES/ INSTALL ELECTRICAL POWER LINES

7.1. Introduction of the Unit of Learning / Unit of Competency

The training of this course unit attracts the skills on the transmission of produce power to the premises of the consumers from the power station. Use of designed and mounted transmission line, adequate amount of power is transmitted to the consumers from the generating stations. Hence the trainee will design and mount the transmission line. The high voltage usage minimizes voltage drop in the line, resistance and reactance and losses during the transmission. 132 kV or 220 kV or 400 kV or 765 kV are among the standard transmission voltages transmitted along the transmission lines are. Therefore, a step-up transformer is required to step up the generated voltage with the aid of erected transmission lines supports at consistent intervals and transmission towers. The line of transmission terminates in a substation where the distribution starts and the distribution lines terminates at load centers.

The trainee will be able to erect the line supports and terminate the cables at different points in the transmission line. The transmission and distribution system are inspected and tested at every stage. While the current's magnitude determines the cost of copper, voltage determines insulator's cost. At load centers voltage levels are stepped down at appropriate values for supply to various consumers. Examples of consumers include both big and medium industries, offices and domestic consumers. Consumers purchase electricity at specific voltage level. For instance, big industries may purchase 132 kV while domestic consumers at lower voltage of 230V, single phase. Therefore, 400 kV transmission voltages should be stepped down before transmitting power to different consumers with the aid of a step-down transformers. Therefore, finally the trainee will inspect and test the transmission lines (Version 2 EE IIT, Kharagpur).

7.2. Performance Standard

Design the transmission line as per the topography of the installation site, the scope of the load, location of the installation, the type of the materials to be used in line installation and the environmental condition of the location in line with the national and international standards, regulations, legislative requirements and in accordance with IEE regulations.

7.3. Learning Outcome

7.3.1. List of Learning Outcomes

1. Designed a transmission lines
2. Erected a transmission lines support
3. Mounted a transmission line
4. Terminated conductors
5. Inspected and Tested installation

7.3.2. Learning Outcome No. 1. Designed a transmission lines

7.3.2.1. Learning Activities

Learning Outcome No. 1. Designed a transmission lines	
Learning Activities	Special Instructions
• Design transmission lines as per the topography (Hilly, Rocky, Swampy, and Flat) of the installation site. This can be through visiting websites, use video tapes, use charts, maps, visit sites and • Use industrial drawing and design type is the transmission line as per the scope of the load. • Design transmission lines as per location of the installation from the provided use video tapes, use charts, maps, visit sites and. • Design the transmission line Span length as per the weight of the conductor, height and type of the support. From a Visit work site, use industrial standards, use video tapes, case study • Design the transmission line in line with the national and international standards(IEE standard, 17th Edition, British Standard KEBS standard).Visit work site, use industrial standards, use video tapes, case study and • Perform design as per the environmental condition of the location in line with the regulations and legislative requirements through visit work site, use industrial standards, use video tapes, case study	

7.3.2.2. Information Sheet No1 Designed a transmission lines

Definition of Terms

Alternating Current (AC) –Electric current whose flows is reversed manner or alternated at 50HZ in Kenya and60Hz in the U.S.

Conductors (Power Lines) –cables that carry electric current

Corona – Electrical breakdown of the air near high voltage conductors into charged particles.

Current –electricity flow or electrons movement through a conductor typically measured in amperes

Direct Current (DC) – Electric current flowing in a continuous and same direction as contrasted with alternating current.

Distribution Line – A line carrying electricity and distribute power from the generating station to the end-use customers.

Electric Load – appliances that that consumes electricity

Electric Power Transmission – The process of transporting electric power over long distances to the consumers

Energy – The amount of work that can be done by electricity measured in kilowatt-hours (kWh) or megawatt-hours (MWh).

Foundation – System involved in transferring to the ground the various dead and live loads of the transmission structure and conductors.

Generation – The production of electric energy.

Insulators – Used to contain, separate, or support electrical conductors.

Interconnection – Points on a grid or network where two or more transmission lines are connected at a substation or switching station, or where one stage of the energy supply chain meets the next.

Load Center – A particular geographical area where energy is used.

Power – Rate at which electricity does work and measured in watts or kilowatts (kW) or megawatts (MW).

Shield and Ground Wire – Wires for protection from lightning strikes and corresponding surges.

Substation – Part electrical transmission system that either steps up or down voltage.

Switching Station – Part of an electrical transmission system involved in tying two or more electric circuits together through switches hence permitting a circuit to be disconnected, or to change the electric connection between circuits.

Transmission Structures – Are vital in keeping high-voltage conductors that are separated from their surroundings and from each other.

Voltage – Electric “pressure” measured in volts.

Watt – Unit of electrical power.

Transmission Line – A line carrying electricity and transmit electric power from the generating station to the consumers.

Types of transmission lines

Classified as per the amount of voltage it carries

- High Voltage – voltage above 66kv
- Low voltage-voltage below 66kv

Classified as per the type of transmission line

- Overhead transmission line
- Underground transmission line

Classification of transmission line as per the length and voltage it carries

- Short:- Transmission line length is up to 50km
 - Voltage is comparatively low, less than 20kv
 - The capacitance effects are small and hence can be neglected.
 - Only resistance and inductance are considered in the design.

- Medium:- Transmission line length is between 50km and 150km
-Voltage is comparatively low, greater than 20kv and less than 100kv
-resistance inductance and capacitance are considered in the design.
- Long:- Transmission line length is over 150km
-Voltage is very high, greater than 100kv
-resistance inductance and capacitance are considered in the design.

The elements of a transmission line include; Conductors, supports, insulators, cross arms, Phase plates, Lightning arresters, danger plates, anti-climbing tools, transformers, Isolators bolts and nuts etc

Transmission line designs;

- Transmission line Mechanical design
- When carrying out the Transmission line Mechanical design we establish;
 - The appropriate conductor size and length as per the line voltage
 - The appropriate conductor material as per the line voltage
 - The appropriate line support as per the line voltage
 - The correct size and number of insulators as per the line voltage
 - Allowable sag as per the line voltage
 - Expected corona losses as per the line voltage
 - The string efficiency as per the line voltage
 - The appropriate sag and tension of the line of transmission as per the line voltage

- Transmission line Electrical design
- When carrying out the electrical design of the line of transmission we establish;
 - The transmission line resistance as per the conductor material.
 - The flux linkages around the transmission line as per the line voltage.
 - The inductance of the line of transmission as per the line voltage.
 - The electric potential around the conductor as per the line voltage.

7.3.2.3. Self-Assessment

1. State any Two transmission line parameters
2. Explain the function of any five elements of a transmission line.
3. Differentiate a long transmission line, medium transmission line and short transmission line.
4. Explain a transmission Line
5. Describe types of transmission line
6. Explain the classification of transmission line as per the length and voltage it carries

7. What are the elements of a transmission line?
8. Develop an understanding of transmission line design
9. transmission line length is up to 50km, Voltage is comparatively low, less than 20kv and the capacitance effects are small and hence can be neglected.
 - a) Short
 - b) medium
 - c) Long
10. transmission line length is between 50km and 150km
 - a) Short
 - b) medium
 - c) Long
11. transmission line length is over 150km, Voltage is very high, greater than 100kv and resistance inductance and capacitance are considered in the design.
 - a) Short
 - b) medium
 - c) Long
12. The elements of a transmission line include the following except;
 - a) Conductors
 - b) Supports
 - c) Contactors
 - d) insulators
13. When carrying out the Transmission line Mechanical design we establish the following except
 - a) The appropriate conductor size and length as per the line voltage
 - b) The appropriate conductor colour as per the line voltage
 - c) The appropriate line support as per the line voltage
 - d) The correct size and number of insulators as per the line voltage
14. The transmission and distribution system are inspected and tested at every stage.
 - a) True
 - b) False

Practical exercise

- a) Demonstrate ability to design transmission lines as per the topography (Hilly, Rocky, Swampy, and Flat) of the installation site. This can be through visiting websites, use video tapes, use charts, maps, visit sites and
- b) see industrial drawing and design type is the transmission line as per the scope of the load.
- c) Demonstrate ability to design transmission lines as per location of the installation from the provided use video tapes, use charts, maps, visit sites and.
- d) Design the transmission line Span length as per the weight of the conductor, height and type of the support. From a Visit work site, use industrial standards, use video tapes, case study

7.3.2.4. Tools, Equipment, Supplies and Materials for the specific learning outcome

1. Cross arm,
2. Transformers
3. Isolators
4. Insulators
5. Danger plates
6. Lightning arrestors
7. Anti-climbing wire
8. Cable
9. Bolts and Nuts

7.3.2.5. References

1. Electric Power Generation, Transmission, and Distribution and Electrical Protective Equipment – Final Rule Updated 2-18-2015 Performance Standard
2. An Introduction to Electric Power Transmission- Western Governors' Association
3. Version 2 EE IIT, Kharagpur, Principles of power system by v.k mehta

7.3.3. Learning Outcome No. 2. Erected a transmission lines support.

7.3.3.1. Learning Activities

Learning Outcome No. 2. Erected a transmission lines support.	
Learning Activities	Special Instructions
• Use geographical maps, visit site, use video tapes erect Supports as per the location's soil profile. • Visit websites, use video tapes, manufacturer's manual sheet, catalogues, actual site visit and erect supports as per the weight of the conductor. • Use manufacturers' catalogues, working drawings, EMCA Act, OSHA, County by-laws and elect supports as per established procedures and standards • Use manufacturers' catalogues, working drawings, EMCA Act, • OSHA, County by-laws erect supports as per the conductors' voltage.	

7.3.2.2. Information Sheet No. 7/LO2

Transmission line support is a structure used to hold the transmission lines above the ground.

Types of supports

- Wooden poles
- Steel poles
- Reinforced concrete poles
- Steel towers

Material used in manufacturing of transmission line supports e.g.

- Concrete
- Steel
- Wood

Erecting of transmission line support :

Factors to consider when erecting line support

- Wind load ✓
- Ice load ✓

The materials required are;

- Stays
- Struts
- Cross-arm ✓
- Tie struts
- The supports ✓
- Insulators ✓
- Bolt ✓
- Nuts. ✓

Application of transmission

- Importing of power from another country
- Transmitting power from the generating station to a substation

7.3.2.3. Self-Assessment

1. Give four types of line supports
2. State four materials used for making line supports
3. Explain the application of the line support
4. State transmission line support
5. Trainee should be able to use geographical maps
6. What is the importance of manufacturers' catalogues?
7. State four types of line support
8. Explain four factors to consider when installing a transmission line
9. Give the applications of transmission line

7.3.2.4. Tools, Equipment, Supplies and Materials for the specific learning outcome

- Stays
- Struts
- Cross-arm
- Tie struts
- The supports
- Insulators
- Bolt
- Nuts.

7.3.2.5. References

1. Electrical power systems by V.K mehta

7.3.4. Learning Outcome No. 3. Mount a transmission line

7.3.4.1. Learning Activities

Learning Outcome No. 3. Mount a transmission line	
Learning Activities	Special Instructions
• Mount transmission lines in consideration of the balance between tension and sag of the conductors. Visit website, use video tapes, IEE regulations, visit sites and • establish Spacing between conductors as per line voltage Use manufacturers' catalogues, working drawings, EMCA Act, OSHA, County by-laws, make site visit, IEE regulations • mount transmission line as per the type of the conductors. Use manufacturers' catalogues, working drawings, EMCA Act, OSHA, County by-laws, make site visit, IEE regulations and • Perform tension of transmission line as per fluctuations of the weather condition. Visit site, use manufacturer's manual, hold discussions, consultations • Cross arms are mounted in line with the spacing between conductors. • mounted conductors as per the types of insulators used. Use manufacturers' catalogues, working drawings, EMCA Act, OSHA, County by-laws, make site visit, IEE regulations and	

7.3.4.2. Information Sheet No. 3/ LO3

Types of transmission line conductors

- Copper
- Aluminum
- Steel cored aluminum
- Galvanized steel
- Cadmium copper

conductor materials used in transmission line

Equipment's used in mounting transmission line

- Screw drivers
- Pliers kit
- Hand Drill and drill bits selection

<https://www.lowes.com/pl/Pliers-plier-sets-Hand-tools-Tools/4294425055>

- Razor knife for cutting rubber cooler hose
- Hacksaw or small pipe cutter for cutting cooler lines
- A drain pan or small container for capturing fluid
- Climbers
- Drilling Machine
- Transmission-towers-and-substations
 - Hydraulic tensioner
 - Hydraulic pulley
 - Hydraulic pulley Transmission line stringing equipment
 - Hydraulic cable tensioner maximum intermittent tension

Tension and sag in transmission line

The sag and tension analysis help in the continuity and quality of electrical services. Increasing the tension of the conductor is beyond the limit, may break it, and the power transmissions of the system may get erupted.

The conductor dip between the two-level supports is referred to as sag as illustrated below. A span is the horizontal distance between two electrical supports as shown below.

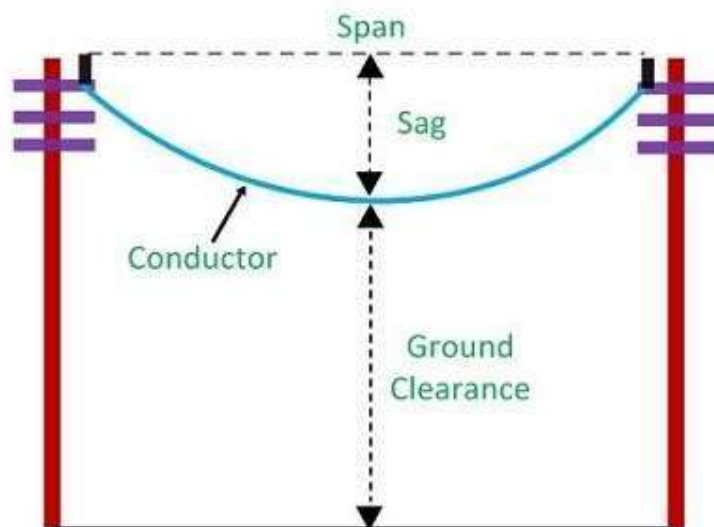


Figure 110: Illustration of Span

Source: <https://circuitglobe.com/calculation-of-sag-and-tension.html>

Factors affecting the sag;

- Conductor weight – the weight of the conductor is directly proportional to the sag. Ice loading increases the weight of the conductor.
- Span – square of the length is directly proportional to the sag. The longer span, the more the sag.
- Tension -The tension in the conductor is inversely proportional to the sag. Increase in tension raises the stress in the insulators that supports the structure.
- Wind –sag is increased in the inclined direction.
- Temperature – At low temperatures, sag is reduced while at high temperatures sag is increased.

<https://circuitglobe.com/calculation-of-sag-and-tension.html>

<https://circuitglobe.com/sag-and-tension.html>

Others include; Insulators, Danger plates, Lightning arrestors, Anti-climbing wire, Cables, Bolts and Nuts

Transmission line spacing and corona effects

II. Power Loss

According to Peek's formulas [5], the power loss due to corona under fair weather conditions can be expressed as:

$$P_{ufwc} = \frac{K}{\delta} (f + 25) \sqrt{\frac{R}{d}} \left(\frac{V_L}{\sqrt{3}} - E_{dcv} \right)^2 L \times 10^{-5} \text{ KW / km / phase} \quad (1)$$

The normalized air density correction factor, δ is 1 at 25°C at 76 cm pressure. Table 1 shows the sample of corona loss calculation based on peek's formula

Under the stormy weather conditions, the disruptive critical voltage (E_{dcv}) is taken as 0.8 times that of the fair weather value. Where the ratio of the phase voltage to the disruptive critical voltage is less than 1.8, Peterson formula [6] is used for determining the corona loss. This is given as:

$$P_{pet} = \frac{21 \times 10^{-6} \times f V^2}{(\log 10 d / R)^2} \quad (2)$$

Power loss due to Corona on High Voltage Transmission Lines (Enesi ,Asizehi), IOSR Journal of Electrical and Electronics Engineering 2013.

Used in transmission line protection e.g.

- Switch gear
- Fuses used in transmission line
- Isolators

- Circuit breakers

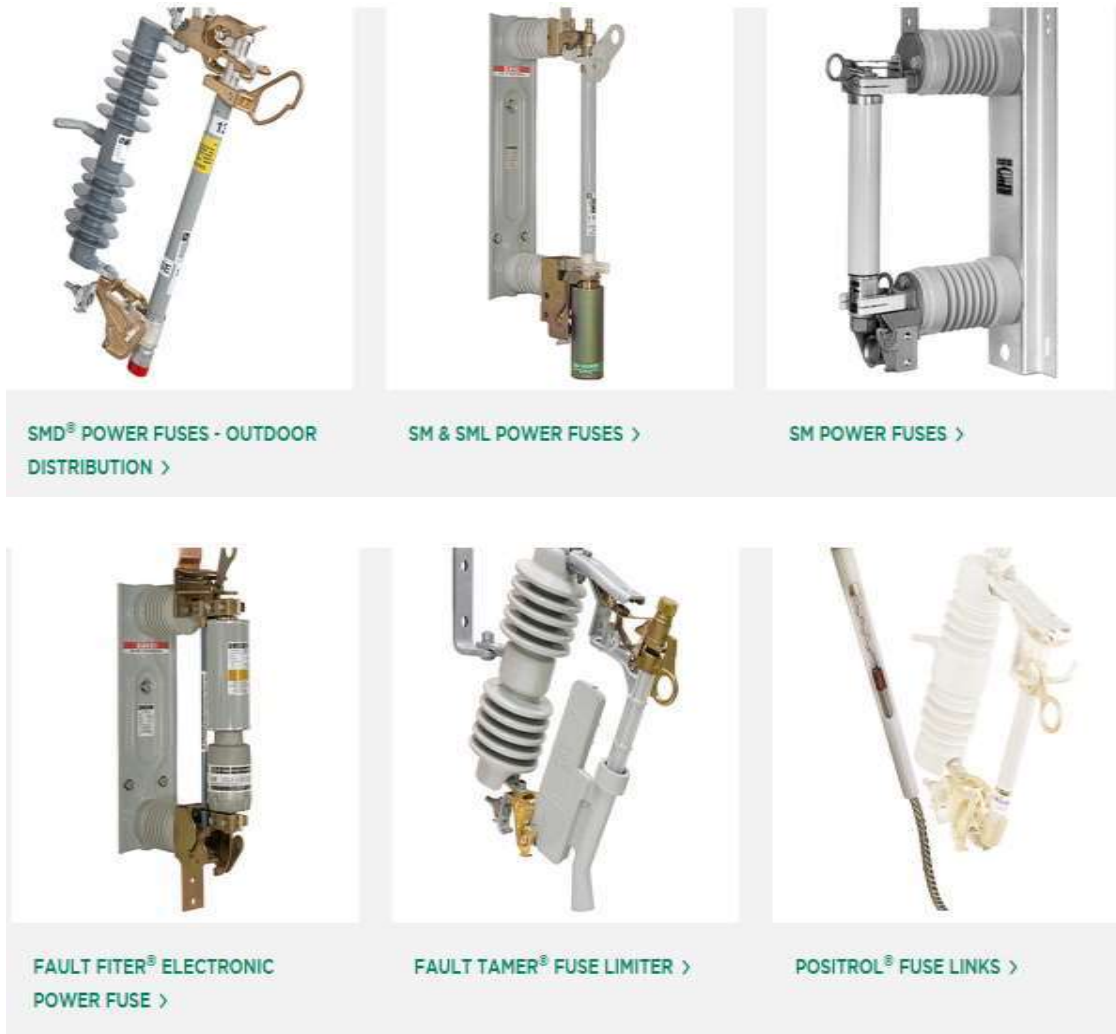


Figure 111: transmission line protection Fuses & Gears

Source: <https://circuitglobe.com/calculation-of-sag-and-tension.html>
<https://circuitglobe.com/sag-and-tension.html>

Definitions

Earthing: Refers to the electrical conduction connection between the construction part and the surrounding soil.

Functional earthing: Earthing to divert lightning currents and 50-periodic fault currents so safe tripping is obtained.

Protective earthing: Earthing a point or points in a system for electrical safety and ensuring protection against electric shock.

Earth conductor

Conductor that forms a part of the fixed installation for the purposes of earthing

Earth electrode: Conductor that provides an electrical connection with the surrounding soil layers.

Shallow earth electrode: Earth electrode that consist of a continuous counterpoise, radial earth electrode or otherwise to the surface parallel located earth conductors down to a depth of about 1 m.

Deep earth electrode: Earth electrode that consist of vertically placed earth electrode buried or driven down in the soil at a greater depth than 1 m.

Radial earth electrode: Horizontally radially placed earth conductors in the soil.

Counterpoise: Conductor or combination of conductors buried in the ground and electrically connecting the footings of the supports of the towers.

Continuous counterpoise: Conductor buried in the ground electrically connecting the footings of the supports of the towers along an overhead line.

Local earth electrode: Earth electrode comprising deep earth electrode, radial earth electrode or a combination thereof.

Protective bonding conductor: Conductor provided for protective-equipotential-bonding.

Potential grading earth electrode: Protective earthing consisting of a protective bonding conductor in the soil around structure and/or stay anchors.

Earthing conductor: Conductor between an earth electrode and an earth terminal or, if such terminal is missing, between an earth electrode and a construction part that shall be earthed.

Down conductor: Conductor for connection of insulator string attachments, cross arm and shield wire to a fixed earth terminal on the tower.

Shield wire: Conductor, horizontally located in an overhead line, usually located above the phase conductors.

Continuous shield wire: Shield wire located along the entire length of overhead lines.

Insulated shield wire: Shield wire electrically separated from the tower with insulators

7.3.2.3 Self-Assessment

1. Differentiate a surge diverter and a surge arrester
2. Explain the parts of a lightning arrester
3. Outline the steps of earthing a transmission line
4. State transmission line conductors
5. Describe types of transmission line conductors
6. Explain tension and sag in transmission line
7. Describe factors affecting the sag
8. Explain components used in mounting transmission lines
9. Perform calculations on power loss
10. Describe transmission line protection
11.consist of vertically placed earth electrode buried or driven down in the soil at a greater depth than 1 m.
 - a) Radial earth electrode

- b) Continuous counterpoise
 - c) Counterpoise
 - d) Deep earth electrode
12.consist of horizontally radially placed earth conductors in the soil.
- a) Radial earth electrode
 - b) Continuous counterpoise
 - c) Counterpoise
 - d) Deep earth electrode
13. Conductor or combination of conductors buried in the ground and electrically connecting the footings of the supports of the towers.
- a) Radial earth electrode
 - b) Continuous counterpoise
 - c) Counterpoise
 - d) Deep earth electrode
14. Inthe conductor buried in the ground electrically connecting the footings of the supports of the towers along an overhead line.
- a) Radial earth electrode
 - b) Continuous counterpoise
 - c) Counterpoise
 - d) Deep earth electrode
15. The tension in the conductor is inversely proportional to the sag
- a) True
 - b) False
16. At low temperatures, sag is reduced while at high temperatures sag is increased.
- a) True
 - b) False

Practical exercise

- a) Demonstrate ability to use geographical maps, visit site, use video tapes erect Supports as per the location's soil profile.
- b) Visit websites, use video tapes, manufacturer's manual sheet, catalogues, actual site visit and erect supports as per the weight of the conductor.
- c) Identify and use manufacturers' catalogues, working drawings, EMCA Act, OSHA, County by-laws and elect supports as per established procedures and standards

7.3.2.4. Tools, Equipment, Supplies and Materials for the specific learning outcome

- Multimeter
- Surge arrester
- Surge diverter
- Transmission lines

7.3.2.5. References

1. http://www.weschler.com/_upload/sitepdfs/techref/gettingdowntoearth.pdf

2. https://dam-assets.fluke.com/s3fs-public/1579160_b_w.pdf).
3. <https://www.quora.com/Why-are-cross-arms-used-in-transmission-lines>
4. <https://www.transformertechnologies.com/>
5. <https://www.southernstatesllc.com/applications/transmission-line-circuit-isolation/>
6. SVENSKA KRAFTNAT SWEDISH NATIONAL GRID-Technical guideline-2015.
7. <https://www.svk.se/siteassets/aktorsportalen/tekniska-riktlinjer/tr05/tr05-13e-earthing-revision-2.pdf>
8. <https://www.sandc.com/en/products--services/products/smd-power-fuses-outdoor-transmission/>
9. https://dam-assets.fluke.com/s3fs-public/1579160_b_w.pdf).
10. https://www.ee.iitb.ac.in/course/~emlab/assets/oc_sc.pdf
11. <https://www.instructables.com/id/Basic-Electric-CONTINUITY-TESTER/>

7.3.5. Learning Outcome No. 4. Terminated conductors

7.3.5.1. Learning Activities

Learning Outcome No. 4. Terminated conductors	
Learning Activities	Special Instructions
• Visit work site, use industrial standards, video tapes, case study , manufacturers' catalogues, working drawings, EMCA Act, • OSHA, County by-laws and terminate conductors as per the line voltage. • Use manufacturers' catalogues, working drawings, EMCA Act, • OSHA, County by-laws, visit work site, use industrial standards, use video tapes, case study and terminate transmission lines as per the type of conductors • Visit work site, use industrial standards, use video tapes, case study, manufacturers' catalogues, working drawings, EMCA Act, • OSHA, County by-laws and use appropriate types of joint terminations. • Visit work site, use industrial standards, use video tapes, case study, manufacturers' catalogues, working drawings, EMCA Act, • OSHA, County by-laws and terminate transmission lines as per the IEE regulation • Visit work site, use industrial standards, use video tapes, case study and terminate transmission line as per end point load. • Visit work site, use industrial standards, video tapes, case study and terminate transmission line as per the type of transmission.	

7.3.5.2. Information Sheet No. 7/ LO4

Transmission line termination refers to putting to an end a transmission line with a device with similar characteristic impedance of the line to prevent signals from being reflected off at the end of the transmission line.

Passive

Passive terminators comprise of a single resistor, though reactive loads may need other passive components for example inductors, capacitors, or transformers.

Active

Active terminators comprise of voltage regulator that ensures voltage used for the terminating resistor(s) at a constant level.

Forced Perfect Termination.

Forced perfect termination (FPT) is applicable on a single ended buses in where diodes remove over and undershoot conditions. There is locking of signal between two actively regulated voltage levels, resulting in superior performance over a standard active terminator. Figure below;

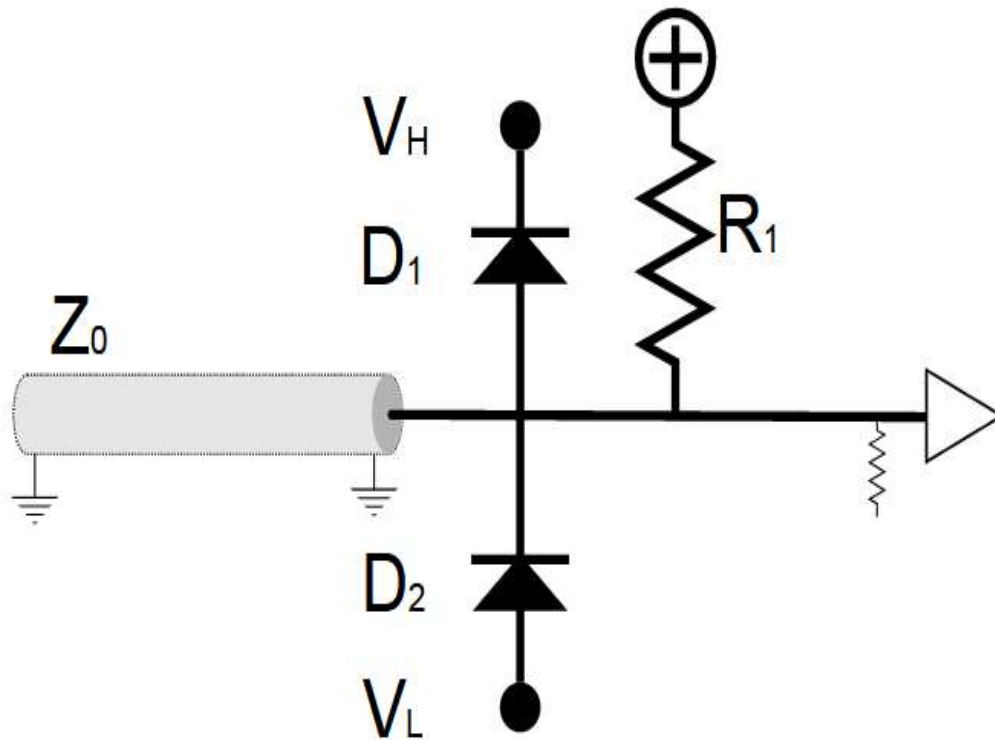


Figure 112: Forced Perfect Termination.

Source: <https://circuitglobe.com/calculation-of-sag-and-tension.html>
<https://circuitglobe.com/sag-and-tension.html>

Some of the termination technique includes:

- Parallel Termination. This technique uses a single resistor from trace to ground and operates appropriately with distributed loads. However, there is a provision of a continuous DC path to ground. Shown below.
- AC Termination. This technique incorporates a capacitor in series with a parallel resistor that terminates to block DC. However, depending on the capacitor used there can be problems with power dissipation or overshoots.

Types of transmission

- i) A.C transmission: It is a transmission line carrying alternating current.
- ii) D.C transmission: It is a transmission line carrying direct current.

Transmission of electric power can be in both AC and DC. They are associated with both advantages and disadvantages as discussed below

DC Transmission

Advantages:

- Only a maximum of two conductors are used in DC transmission -There are no Inductance and Surges in DC transmission.
- Low voltage drop due to absence of inductance
- No skin effect in DC transmission hence only a small cross-sectional area of the conductor is required.
- DC line requires less insulation due to less potential stress.
- No interference with system of communication
- Low Corona losses
- No dielectric losses in high DC transmission.
- No difficulties in synchronizing and stability problems.
- Low insulation required for the Dc systems about 70%).

Disadvantages

- Not possible to produce power at High (DC) Voltage due commutation.
- Not possible to step the level of DC Voltage at high voltage.
- It is not possible to change the level of DC Voltage hence one cannot get desired voltage for Electrical and electronics appliances direct from Transmission system.

AC Transmission

Advantages:

- AC Circuit breakers are cheap
- Easy maintenance and repair of AC sub-station
- It is possible to either increase or decrease AC voltage using transformer

Disadvantages:

- Has skin effect, hence results to losses during transmission.
- Power loss due to capacitance and inductance. along the Corona Losses occur in AC lines,
- There is telecommunication interference in AC System.
- There are problems with stability and synchronization.

Type of loads

Electrical loads such as bug zappers, toasters and refrigerators are the devices connected to the power system. The three types of electrical loads discussed below vary depending on the leading and lagging time relationship between the current and the voltage.

The three load types are

- Resistive
- Inductive
- Capacitive

Components used in line termination e.g.

- Lugs
- Fuses
- Droppers

7.3.5.3. Self-Assessment

1. state four types of electrical joints
2. state the type of electrical load and give an example
3. State transmission line termination
4. Describe types of transmission line termination
5. Explain termination techniques
6. What is the importance of termination?
7. Differentiate between AC and DC transmission
8. Which of the following isn't an advantage of AC transmission?
 - a) AC Circuit breakers are cheap
 - b) Easy maintenance and repair of AC sub-station
 - c) No difficulties in synchronizing and stability problems.
 - d) It is possible to either increase or decrease AC voltage using transformer
9. The three load types include the following except
 - a) Resistive
 - b) Inductive
 - c) Terminative
 - d) Capacitive
10. Components used in line termination include the following except
 - a. Lugs
 - b. Fuses
 - c. Droppers
 - d. Transformer
11. Active terminators comprise of voltage regulator that ensures voltage used for the terminating resistor(s) at a constant level.
 - a) True
 - b) False

12. Thevenin Termination. This technique uses resistor pairs, one for ground, the other for power.

- a) True
- b) False

Practical exercise

- a) Demonstrate ability to use manufacturers' catalogues, working drawings, EMCA Act, OSHA, County by-laws, visit work site, use industrial standards, use video tapes, case study and terminate transmission lines as per the type of conductors
- b) Visit work site, use industrial standards, use video tapes, case study, manufacturers' catalogues, working drawings, EMCA Act, OSHA, County by-laws and use appropriate types of joint terminations.

7.3.5.4 Tools, Equipment, Supplies and Materials for the specific learning outcome

- DC generator
- AC generator
- Multimeter
- Electrical cables
- Soldering iron

7.3.5.5. References

- 1. <https://www.youtube.com/watch?v=puF21Z2hfIM>
- 2. <http://electriciantraining.tpub.com/14182/css/Terminating-A-Transmission-Line-146.htm>
- 3. <https://www.falstad.com/circuit/e-tlterm.html>https://en.wikipedia.org/wiki/Electrical_termination
- 4. https://en.wikipedia.org/wiki/Electrical_termination<https://fenix.tecnico.ulisboa.pt/downloadFile/3779579943588/>

7.3.6. Learning Outcome No. 5. Inspected and Tested installation

7.3.6.1. Learning Activities

Learning Outcome No. 5. Inspected and Tested installation	
Learning Activities	Special Instructions
• Visit work site, use industrial standards, use video tapes, case study, manufacturers' catalogues, working drawings, EMCA Act, OSHA, County by-laws and establish the tests to be carried. • Visit work site, use industrial standards, use video tapes, case study, manufacturers' catalogues, working drawings, EMCA Act, OSHA, County by-laws and perform the test as per the IEE regulations. • Visit work site, use industrial standards, use video tapes, case study, manufacturers' catalogues, working drawings, EMCA Act, OSHA, County by-laws and establish the firmness of the transmission line. • Visit work site, use industrial standards, use video tapes, case study, manufacturers' catalogues, working drawings, EMCA Act, OSHA, County by-laws and perform Continuity test. • Visit work site, use industrial standards, use video tapes, case study, manufacturers' catalogues, working drawings, EMCA Act, OSHA, County by-laws and perform Short circuit test. • Visit work site, use industrial standards, use video tapes, case study, manufacturers' catalogues, working drawings, EMCA Act, OSHA, County by-laws and perform Insulation test as per the IEE regulations	

7.3.6.2. Information Sheet No. 7/ LO5

- Definitions of key terms
- methods, processes/ procedures/ guidelines
- content
- illustrations (photographs, pictures, videos, charts, plans, digital content, simulations)
- case studies

Meaning of testing in line transmission

Testing means using instrument to check the proper operation of the transmission line.

Types of tests in line transmission e.g.

- Continuity test- <https://www.instructables.com/id/Basic-Electric-Continuity-Tester/>

How to test for continuity

1. Turn the dial to Continuity Test mode. It will likely share a spot on the dial with one or more functions, usually resistance (Ω). With the test probes separated, the multimeter's display may show OL and Ω .
2. If required, press the continuity button.
3. First insert the black test lead into the COM jack.
4. Then insert the red lead into the V Ω jack. When finished, remove the leads in reverse order: red first, then black.
5. With the circuit de-energized, connect the test leads across the component being tested. The position of the test leads is arbitrary. Note that the component may need to be isolated from other components in the circuit.
6. The digital multimeter (DMM) beeps if a complete path (continuity) is detected. If the circuit is open (the switch is in the OFF position), the DMM will not beep.
7. When finished, turn the multimeter OFF to preserve battery life.

Proper insulation is associated with high resistance and poor insulation

The actual values of resistance can either be low or high depending on factors including temperature or the insulation moisture content.

The Megger insulation tester is vital in testing insulation resistance measured in ohms or megohms. It is usually a high-range resistance meter (ohmmeter) with a built-in direct-current generator. It has special current and voltage coils to enable true ohms to be read directly, independent of the actual voltage applied. This method is non-destructive hence does not deteriorate the insulation (https://dam-assets.fluke.com/s3fs-public/1579160_b_w.pdf).

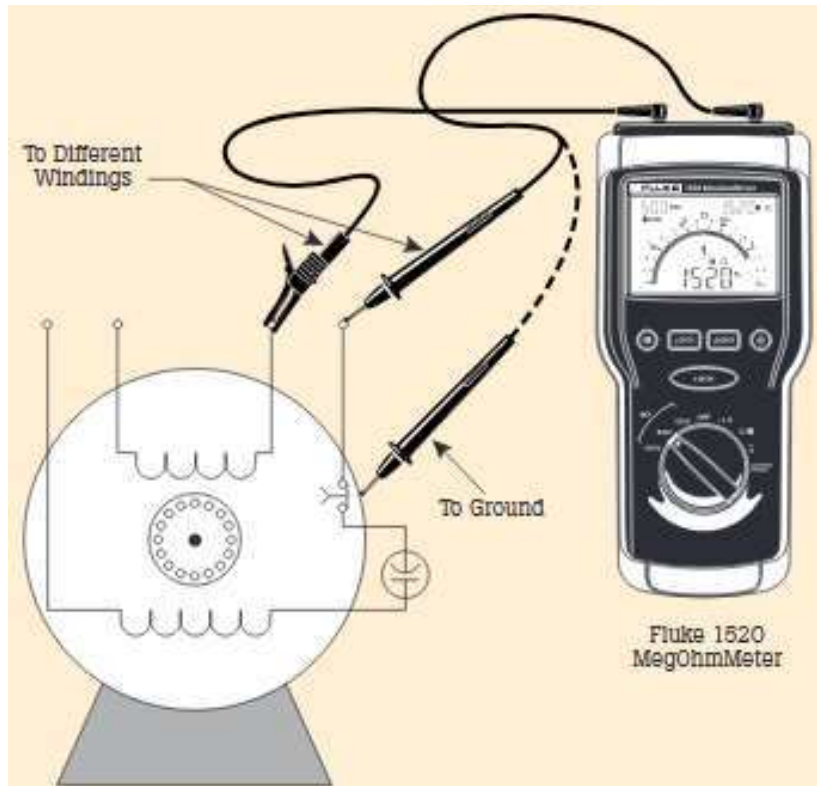


Figure 113: Insulation resistance tester carrying out measurement(https://dam-assets.fluke.com/s3fs-public/1579160_b_w.pdf).

Earth continuity test

Below are drawings showing how to measure earth resistance.

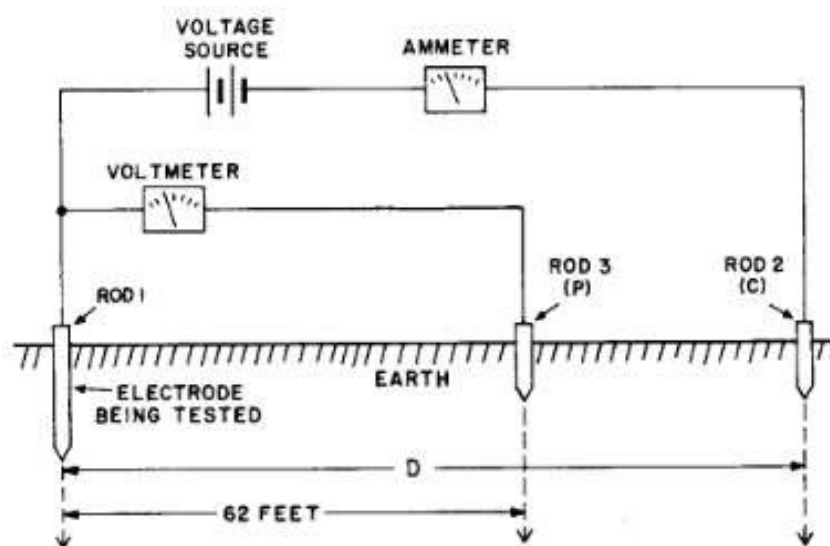


Figure 114: measuring earth resistance
(http://www.weschler.com/_upload/sitepdfs/techref/gettingdowntoearth.pdf)

7.3.6.3 Self-Assessment

- 1) State testing in line transmission
- 2) What is the importance of testing in line transmission?
- 3) Describe types of tests in line transmission
- 4) Explain how to test for continuity
- 5) In the actual values of resistance can either be low or high depending on factors including temperature or the insulation moisture content.
 - a) Continuity test
 - b) Insulation test
 - c) Earth continuity test
- 6) In the digital multimeter (DMM) beeps if a complete path (continuity) is detected.
 - a) Continuity test
 - b) Insulation test
 - c) Earth continuity test
- 7) Which of the following is not a type of test in line transmission?
 - a) Continuity test
 - b) Insulation test
 - c) Earth continuity test
 - d) Speed test
- 8) The Megger insulation tester is vital in testing insulation resistance measured in ohms or megohms.
 - a) True
 - b) False
- 9) Testing means using instrument to check the proper operation of the transmission line.
 - a) True
 - b) False

Practical exercise

- a) Identify and demonstrate how to test continuity
- b) Identify and outline the procedure of carrying out the following tests
 - i) Short circuit test
 - ii) Insulation resistance test

7.3.6.4. Tools, Equipment, Supplies and Materials for the specific learning outcome

- Insulation resistance test
- Earth resistance test
- Transmission line
- Continuity tester

7.3.6.5 References (APA)

1. http://www.weschler.com/_upload/sitepdfs/techref/gettingdowntoearth.pdf
2. https://dam-assets.fluke.com/s3fs-public/1579160_b_w.pdf.
3. https://dam-assets.fluke.com/s3fs-public/1579160_b_w.pdf.
4. https://www.ee.iitb.ac.in/course/~emlab/assets/oc_sc.pdf
5. <https://www.instructables.com/id/Basic-Electric-CONTINUITY-TESTER/>